# Anikait Bansal

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## PROFESSIONAL SUMMARY

Technical engineering student specializing in network infrastructure, backend systems, and cybersecurity. Proven experience building complex virtualized SD-WAN architectures, cryptographic overlays, and AI-integrated active-defense honeypots. Seeking to leverage hands-on expertise in Python, Linux networking, routing protocols, and systems architecture for engineering internships.

## EDUCATION

### Dr. B.R. Ambedkar National Institute of Technology (NIT) Jalandhar

Bachelor of Technology in Electrical Engineering | Expected Graduation: 2029

- **CGPA:** 7.76
- **Relevant Coursework:** Multivariable Calculus, Electrical Engineering Systems, Environmental Science (IDFC0101).

## TECHNICAL SKILLS

- **Networking:** Cisco IOS, Open vSwitch (OVS), OSPF, BGP, MPLS, IPsec (IKEv2/AES-256), SNMP, IP SLA, PBR, GNS3, Linux networking (tc/netem).
- **Cybersecurity:** Honeypot Deployment, Active Defense, Threat Intelligence, Web Vulnerability Scanning (SQLi/XSS).
- **Programming & Backend:** Python, FastAPI, Bash/Shell, SQLite, REST APIs, Uvicorn, Pytest.
- **Hardware/Sensors:** LiDAR, Eddy Current Sensors, Accelerometers.

## TECHNICAL PROJECTS

### Enterprise SD-WAN Fault-Injection Testbed

*GNS3, Cisco IOS, BGP, MPLS, IPsec, Python, Linux*

- Architected a 27-node, 3-site enterprise SD-WAN testbed in GNS3 using Cisco IOSv and Open vSwitch to validate dynamic path steering and network resilience.

- Designed a dual-transport architecture featuring a primary MPLS/BGP core and a secondary AES-256-GCM encrypted IPsec overlay for secure routing failover.
- Engineered custom Python UDP traffic generators and Linux tc/netem fault-injection scripts to benchmark sub-second IP SLA failover and Policy-Based Routing (PBR).
- Configured SNMP telemetry polling across OVS trunk switches to capture interface statistics and packet sequence metrics for machine learning anomaly detection.

### DECEPTRA: Honeypot-as-a-Service

*Python, FastAPI, SQLite, LLMs (Ollama), REST API*

- Developed an active-defense framework in Python using FastAPI to intercept, log, and analyze automated bot traffic and malicious network payloads.
- Integrated Large Language Models (Ollama) to dynamically generate mock interactive environments, such as fake database schemas, trapping attackers in deception loops.
- Built a comprehensive REST API and threat intelligence dashboard to monitor attacker IPs, user-agents, raw HTTP headers, and SQL injection attempts in real time.
- Programmed automated bot simulation scripts and implemented a deterministic threat analysis engine, strictly validated by a Pytest suite.

### Industrial Pipeline Crawler Architecture

*Sensor Networks, Robotics*

- Architected the technical workflow for an industrial-grade sewage and gas pipeline crawler designed to map complex underground infrastructures.
- Integrated theoretical models for LiDAR, eddy current sensors, and accelerometers to optimize structural defect detection.

## ACHIEVEMENTS & EXTRACURRICULARS

- **2nd Runner Up:** NIT Jalandhar Buildathon.
- **Participant:** ECLIPSE Hackathon at THAPAR Patiala.
- **Research:** Drafted a comprehensive technical report analyzing the Buddha Nullah water system ecology under Dr. N.C. Kothiyal.